

Hands-on Network & Service Checklist

1. Identity

Command: hostname -I

```
ubuntu@ip-172-31-17-115:~$ hostname -I
172.31.17.115
ubuntu@ip-172-31-17-115:~$
```

Note your IP address, e.g., 172.31.17.115

Command: ip addr show

```
ubuntu@ip-172-31-17-115:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 06:2a:9d:4e:8c:29 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname enx062a9d4e8c29
    inet 172.31.17.115/20 metric 100 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 3497sec preferred_lft 3497sec
    inet6 fe80::42a:9dff:fe4e:8c29/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
ubuntu@ip-172-31-17-115:~$
```

Alternatively, view detailed network info.

2. Reachability

Command: ping <target>

```
ubuntu@ip-172-31-17-115:~$ ping google.com
PING google.com (142.251.142.238) 56(84) bytes of data.
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=1 ttl=118 time=3.71 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=2 ttl=118 time=3.73 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=3 ttl=118 time=3.73 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=4 ttl=118 time=3.74 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=5 ttl=118 time=3.73 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=6 ttl=118 time=3.74 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=7 ttl=118 time=3.74 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=8 ttl=118 time=3.74 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=9 ttl=118 time=3.78 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=10 ttl=118 time=3.74 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=11 ttl=118 time=3.77 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=12 ttl=118 time=3.76 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=13 ttl=118 time=3.74 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=14 ttl=118 time=3.76 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=15 ttl=118 time=3.74 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=16 ttl=118 time=3.76 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=17 ttl=118 time=3.73 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=18 ttl=118 time=3.75 ms
64 bytes from lcarbn-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=19 ttl=118 time=3.74 ms
^C
--- google.com ping statistics ---
19 packets transmitted, 19 received, 0% packet loss, time 18034ms
rtt min/avg/max/mdev = 3.706/3.742/3.777/0.015 ms
ubuntu@ip-172-31-17-115:~$
```

Example: ping google.com

Observe latency (ms) and packet loss (if any).

Observation: Typical latency < 100ms; packet loss indicates network issues if > 0%.

3. Path

Installing tarceroute

```
ubuntu@ip-172-31-17-115:~$ sudo apt install traceroute
Installing:
  traceroute

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 39
  Download size: 54.7 kB
  Space needed: 162 kB / 4662 MB available

Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 traceroute amd64 1:2.1.6-1build1 [54.7 kB]
Fetched 54.7 kB in 0s (4288 kB/s)
Selecting previously unselected package traceroute.
(Reading database ... 85301 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.6-1build1_amd64v3.deb ...
Unpacking traceroute (1:2.1.6-1build1) ...
Setting up traceroute (1:2.1.6-1build1) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
```

Command: `traceroute <target>`

Or: `tracpath <target>`

Example: `traceroute google.com`

```
ubuntu@ip-172-31-17-115:~$ traceroute google.com
traceroute to google.com (142.251.142.238), 30 hops max, 60 byte packets
 1  242.8.132.135 (242.8.132.135)  3.891 ms 242.8.132.131 (242.8.132.131)  4.283 ms 242.8.133.7 (242.8.133.7)  3.853 ms
 2  * * *
 3  173.194.124.194 (173.194.124.194)  3.614 ms 3.598 ms *
 4  142.250.236.117 (142.250.236.117)  4.156 ms 4.145 ms 4.737 ms
 5  142.251.236.73 (142.251.236.73)  4.353 ms 142.251.236.71 (142.251.236.71)  4.108 ms 4.098 ms
 6  lcarbn-ah-in-f14.1e100.net (142.251.142.238)  3.818 ms 3.695 ms 3.804 ms
ubuntu@ip-172-31-17-115:~$
```

Note any long hops (> 30ms) or timeout (* * *) indicating delays or issues.

Observation: Several hops with high latency or timeouts(* * *) suggest routing issues.

4. Ports

Installing netstat

```
ubuntu@ip-172-31-17-115:~$ sudo apt install net-tools
Installing:
  net-tools

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 39
  Download size: 152 kB
  Space needed: 680 kB / 4662 MB available

Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 net-tools amd64 2.10-2ubuntu1 [152 kB]
Fetched 152 kB in 0s (8928 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 85321 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-2ubuntu1_amd64v3.deb ...
Unpacking net-tools (2.10-2ubuntu1) ...
Setting up net-tools (2.10-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes... [=====]
```

Command: ss -tulpn

```
ubuntu@ip-172-31-17-115:~$ ss -tulpn
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port      Process
udp        UNCONN     0           0           127.0.0.54:53            0.0.0.0:*
udp        UNCONN     0           0           127.0.0.53%lo:53         0.0.0.0:*
udp        UNCONN     0           0           172.31.17.115%ens5:68    0.0.0.0:*
udp        UNCONN     0           0           127.0.0.1:323            0.0.0.0:*
udp        UNCONN     0           0           [::1]:323                [::]:*
tcp        LISTEN     0           4096         127.0.0.54:53            0.0.0.0:*
tcp        LISTEN     0           4096         0.0.0.0:22                0.0.0.0:*
tcp        LISTEN     0           4096         127.0.0.53%lo:53         0.0.0.0:*
tcp        LISTEN     0           4096         [::]:22                   [::]:*
```

Lists active listening services and ports.

Observation: LISTEN 0 :: *:22 indicates :The service (probably SSH daemon sshd) is listening for incoming SSH connections on all IPv6 network interfaces on port 22.

5. Name Resolution

Command:

dig <domain>

Example: dig google.com

```

ubuntu@ip-172-31-17-115:~$ dig google.com

; <<>> DiG 9.20.18-1ubuntu2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25096
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                261     IN      A      142.251.142.238

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri May 29 05:20:52 UTC 2026
;; MSG SIZE rcvd: 55

ubuntu@ip-172-31-17-115:~$

```

Or: nslookup [google.com](https://www.google.com)

```

ubuntu@ip-172-31-17-115:~$ nslookup google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.142.238
Name:   google.com
Address: 2a00:1450:400f:809::200e

ubuntu@ip-172-31-17-115:~$

```

Observation: resolved IP, e.g., 142.251.142.238

6. HTTP Check

Command: curl -I http://<target>

Example: curl -I [http://google.com](https://www.google.com)

```

ubuntu@ip-172-31-17-115:~$ curl -I https://google.com
HTTP/2 301
location: https://www.google.com/
content-type: text/html; charset=UTF-8
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-d9cldUj36nfQrTFv6GHiXw' 'strict-dynamic' 'report
https: http;;report-uri https://csp.withgoogle.com/csp/gws/other-hp
date: Fri, 29 May 2026 05:25:24 GMT
expires: Sun, 28 Jun 2026 05:25:24 GMT
cache-control: public, max-age=2592000
server: gws
content-length: 220
x-xss-protection: 0
x-frame-options: SAMEORIGIN
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

ubuntu@ip-172-31-17-115:~$ curl -I http://google.com
HTTP/1.1 301 Moved Permanently
Location: https://www.google.com/
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-xo1EFHnzXHdjOVZi7Q0slg' 'strict-dynamic' 'report
https: http;;report-uri https://csp.withgoogle.com/csp/gws/other-hp
Date: Fri, 29 May 2026 05:28:06 GMT
Expires: Sun, 28 Jun 2026 05:28:06 GMT
Cache-Control: public, max-age=2592000
Server: gws
Content-Length: 219
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN

ubuntu@ip-172-31-17-115:~$ █

```

Note the HTTP status code, e.g., 200 OK.

Observation: Status 200 indicates success; 404 or 500 signals issues.

My observations: The status code 301 indicates that the request has been moved permanently to the URL <https://www.google.com/>. Your client (curl) is being informed to redirect to the new location.

7. Connections Snapshot

Command: netstat -an | head

Count ESTABLISHED vs LISTEN states.

```

ubuntu@ip-172-31-17-115:~$ netstat -an|head
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.0.54:53         0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0.22             0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.0.53:53         0.0.0.0:*               LISTEN
tcp        0      0 172.31.17.115:22        13.48.4.202:21875       ESTABLISHED
tcp6       0      0 :::22                  :::*                     LISTEN
udp        0      0 127.0.0.0.54:53         0.0.0.0:*               *
udp        0      0 127.0.0.0.53:53         0.0.0.0:*               *
udp        0      0 172.31.17.115:68        0.0.0.0:*               *
ubuntu@ip-172-31-17-115:~$ █

```

Observation: 1 ESTABLISHED, 4 LISTEN, indicating active connections.

Mini Task: Port Probe & Interpret

Identify one listening port from ss -tulpn (e.g., SSH on 22 or a local web app).

From the same machine, test it: nc -zv localhost <port> (or curl -I http://localhost:<port>).

Write one line: is it reachable? If not, what's the next check? (e.g., service status, firewall).

Step 1: Identify a Listening Port

Run the command to list open ports and services:

```
ss -tulpn
```

Look for a line with LISTEN in the State column, and note the port number.

Example output:

	Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
tcp	LISTEN	0	128	0.0.0.0:22	0.0.0.0:*		sshd

Here, port 22 is listening (SSH).

Step 2: Test Connectivity to the Port

Use nc (netcat) to check if the port is reachable locally:

```
nc -zv localhost 22
```

Or, test with curl if it's a web port (like 80 or 443):

```
curl -I http://localhost:80
```

Step 3: Interpret the Result

If output shows "succeeded" or "open":

```
Connection to localhost 22 port [tcp/ssh] succeeded!  
=> The port is reachable.
```

If output shows "refused" or "timed out":

```
nc: connect to localhost port 22 (tcp) failed: Connection refused  
=> The port is not reachable.
```

Step 4: Next Checks if Not Reachable

Check if the service is running:

```
sudo systemctl status sshd
```

Check firewall rules:

```
sudo iptables -L -n  
Or, if using ufw:
```

```
sudo ufw status
```

Ensure the port is allowed through the firewall.

Which command gives you the fastest signal when something is broken?

What layer (OSI/TCP-IP) would you inspect next if DNS fails? If HTTP 500 shows up?

Two follow-up checks you'd run in a real incident.

1. Fastest command for quick signal when something is broken:

ping — sends ICMP echo requests to a target host and quickly indicates if the host is reachable and responsive.

```
ping -c 4 <hostname or IP>
```

Why: It's fast, simple, and provides immediate feedback on network connectivity.

2. Layer to inspect if DNS fails:

Next layer: Application Layer (Layer 7 of OSI / Layer 4 of TCP/IP), specifically the DNS service itself.

In terms of protocol: DNS (Domain Name System).

What to check:

DNS server configuration (systemctl status named or systemctl status systemd-resolved)

DNS query with dig or nslookup:

```
dig google.com
or
nslookup google.com
```

3. Layer to inspect if HTTP 500 error appears:

Next layer: Application Layer (Layer 7) — the web server or application hosting the service.

What to check:

Web server logs (e.g., Apache/Nginx logs)

Application logs for errors

Backend services or database connectivity (if applicable)

4. Two follow-up checks in a real incident:

Check service status and logs:

For the web server or relevant service:

```
sudo systemctl status nginx  
sudo journalctl -u nginx
```

Test backend connectivity:

Verify database or internal service health, e.g., via telnet or specific API calls.

Example:

```
telnet 127.0.0.1 3306 # for MySQL
```